

AI and Computer Vision-Based Waterlogging Detection and Alert System

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Abstract— Rooftop waterlogging monitoring is now crucial for building maintenance due to its frequent occurrence caused by drain blockages. Early and accurate detection of stagnant water has now become crucial to reduce structural damage and leakage impacts in residential and commercial buildings. This project explores the adaptability of deep learning techniques, specifically the InceptionV3 model, for rooftop water accumulation detection. This study utilizes the InceptionV3 architecture, selected after considering the capabilities of deep learning models such as VGG16, Resnet50, and MobileNet and their performance for water detection tasks. InceptionV3 is chosen for its effectiveness in transfer learning with extracted features from pre-trained ImageNet layers. The model architecture was modified by including global average pooling and a dropout layer with a dense output layer. Fine-tuning is performed in its last few layers to specifically train the rooftop drainage images. This project demonstrates the potential use of deep learning models in preventative infrastructure maintenance tasks like waterlogging detection. It also demonstrates the power of a deep learning model like InceptionV3 in image recognition and classification tasks with transfer learning techniques, aiming for exceptional performance comparable to state-of-the-art benchmarks in accuracy, F1 score, and precision-recall scores. This blockage detection system has the potential for real-time deployment, analyzing images from mounted surveillance cameras. Real-time water detection can trigger automatic alerts, notifying building owners and maintenance personnel before damage occurs. This is particularly relevant considering the frequent post-rain pooling events on flat rooftops. A practical application involves integrating the model with existing security camera systems. Upon detecting water accumulation, the system utilizes automated email or SMS services to alert stakeholders, enabling them to clear blockages promptly. This project presents a valuable approach to improving automated building management, particularly drain blockage and waterlogging detection.

Keywords— *Rooftop Waterlogging Detection, InceptionV3, Transfer Learning, Deep Learning, Drain Blockage Monitoring*

I. INTRODUCTION

Modern urban infrastructure is highly susceptible to water damage, compounded by architectural designs featuring flat rooftops, dense building layouts, and unpredictable weather

patterns. Structural issues like roof ponding, leakages, mold growth, and concrete degradation present a major risk to both building longevity and property value globally. In the absence of adequate monitoring and efficient maintenance strategies, these structural hazards frequently result in catastrophic events that lead to substantial economic losses, safety hazards, and destruction of building interiors[1]. Situated in environments facing increasingly erratic weather, buildings face annual challenges characterized by heavy rainfall, resulting in sudden water accumulation that pools on rooftops without adequate warning. These waterlogging events stem from various factors, including intense rainfall, debris accumulation, clogged drainage holes, and poor routine maintenance. The inaccessibility of many rooftop areas exacerbates the impact, leading to delayed detection, structural weakening, and disruption of daily operations.

The pervasive computing technology proves to be beneficial in these domains as it leverages context to efficiently handle pertinent information and services. Addressing rooftop water accumulation requires a comprehensive approach integrating scientific knowledge, technological innovations, automated monitoring, and proactive maintenance policies. This involves investing in early warning systems, bolstering blockage detection capabilities, improving infrastructure resilience, and promoting smart building practices. Furthermore, leveraging advanced technologies like Internet of Things (IoT), smart surveillance, and Artificial Intelligence (AI) can significantly improve drainage monitoring, detection, and management. By harnessing data analytics and real-time information, facility managers can make informed decisions, allocate maintenance resources efficiently, and minimize the impact of water damage on structures. Considerable efforts have been undertaken worldwide to create affordable and reliable infrastructure monitoring solutions. One widely adopted method involves utilizing computer vision technology, which involves capturing and analysing pertinent images from mounted

security cameras. This approach aims to enhance the accuracy of maintenance-related decision-making processes .

Commercial and residential areas drive economic growth and innovation. However, this rapid construction comes at a cost, with one of the most pressing challenges being the increased risk of structural deterioration due to water. As urban landscapes expand, flat roof designs become standard for utility placement, yet they are prone to drainage failures. Debris such as leaves, dust, and trash often obstruct drain guards, reducing the system's ability to channel water. This leads to increased surface pooling, overwhelming existing drainage systems and causing stagnant water that can devastate roof integrity. The frequency and severity of these events are further exacerbated by climate change. Global temperature rise is highly effective, including intense rainfall bursts. These changes create a perfect storm for increased rooftop waterlogging risks. The consequences of undetected waterlogging are extensive and destructive. Given the growing threat of structural damage, developing effective methods for early detection is crucial. Early warnings allow maintenance teams to take timely mitigation measures, such as clearing blockages, inspecting drains, and preventing leaks. Building owners can also be alerted to take necessary precautions to protect their assets. Traditional leak detection methods often rely on manual inspections or invasive moisture sensors. While these methods provide valuable information, they have limitations. Manual inspections are labor-intensive and dangerous during storms, and sensors may not capture the visual precursors of blockage caused by debris accumulation. Additionally, the time it takes for dampness to permeate the structure can delay the issuance of early warnings. The significance of enhanced rooftop monitoring cannot be overstated in light of the escalating uncertainty linked to climate change and the rising cost of building repairs [9].

In waterlogging detection, the transition from relying solely on manual checks or complex sensor networks to more localized and visual methodologies represents a significant evolution in approach. While sensors are effective for moisture detection, they cannot often capture the visual details necessary for evaluating the immediate cause of blockage on specific drains. This gap has sparked interest in visual detection techniques, which utilize video surveillance and image processing to accurately determine water levels and blockage status. Computer vision relies on the utilization of image processing methods that have found extensive applications across various domains such as aerospace, medicine, traffic surveillance, and environmental object analysis [5]. The key contributions of this research are:

- Enhanced Rooftop waterlogging detection system by utilizing a modified InceptionV3 model, significantly improving the accuracy and reliability of identifying and classifying drain blockages and water pooling.
- Integrate Artificial intelligence and computer vision methodologies to transform traditional building maintenance approaches and real-time drainage monitoring.

- The proposed model represents a scalable and cost-effective solution for an automated flood monitoring system using standard surveillance cameras.

The paper is structured as follows: Section 2 provides related works, discussing previous research and methodologies in water detection. This is followed by a description of the proposed methodology in Section 3, including the modified InceptionV3 model and its implementation. The results and discussion Section 4 presents the findings and compares them with existing methods. Finally, Section 5 summarizes the conclusion and contributions and suggests directions for future research.

II. RELATED WORKS

In the context of water detection, numerous researchers have executed different techniques to attain satisfactory performances using AI and Deep Learning models. Traditional methods often lag in speed and accuracy, prompting a shift toward advanced technologies like computer vision.

Computer Vision and Deep Learning Approaches Huang et al. [10] presented a method to spot floods quickly using computer vision, testing algorithms on real-time data. They found that while EfficientNet performed best with over 99.5% accuracy, 3D CNNs struggled with dynamic backgrounds—a challenge also relevant to rooftop environments with changing weather conditions. They boosted accuracy using ResNet and image augmentation, achieving a 99.5% recognition rate. Similarly, Zhong et al. [23] integrated image recognition to identify submerged objects (like vehicles) to estimate flood depth, achieving 89.29% accuracy using YOLOv4. While effective for streets, these methods require reference objects often absent on flat rooftops. Our project adapts these visual principles, using the drain itself as the reference object to identify blockage-induced pooling.

IoT and Sensor-Based Monitoring Pathan et al. [2] addressed flood issues using IoT, specifically suggesting the deployment of Arduino Uno for detection and Raspberry Pi for processing. This validates our choice of using similar cost-effective hardware for building maintenance. However, they noted limitations in sensor data accuracy and scalability. Novianti et al. [3] advocated for using CCTV and Deep Learning for water level assessment to create geospatial models. While innovative, they highlighted that visibility issues and maintenance of camera systems remain significant challenges. Our project aims to overcome these sensor limitations by relying on direct visual classification of the drain state rather than complex depth measurements.

Limitations of Rainfall and Numerical Models Zhang et al. [4] introduced a model to predict waterlogging depths based solely on historical meteorological data and rainfall patterns. While their model surpassed traditional methods for short-term events, it solely relies on rainfall data, neglecting factors like urban infrastructure conditions. Crucially, a rainfall model cannot detect if a specific rooftop drain is physically blocked by debris. Similarly, Chen et al. [5] combined numerical models with LSTM neural networks for risk prediction. While efficient for city-scale forecasting, these models struggle with data scarcity in complex terrains and cannot provide the granular, real-time "blocked/unblocked" status required for immediate building maintenance.

The Need for Specialized Datasets A significant hurdle in this domain is data availability. Khouakhi et al. [6] emphasized the importance of flood-related image datasets from diverse sources like surveillance cameras. They highlighted that the lack of openly accessible and labeled flood image datasets hinders the progress of CNN applications. This scarcity is even more pronounced for specific infrastructure scenarios like rooftop drainage. Just as Alizadeh and Behzadan [7] utilized crowdsourced images of traffic signs to estimate flood depth, our research addresses this gap by creating a specialized dataset for rooftop drain blockages, enabling the specific application of Transfer Learning models like InceptionV3 where general datasets fail.

III. METHODOLOGY

Unlike determining urban floods from public repositories, our dataset was constructed using a custom camera module mounted at the surveillance site. A total of 450 images were collected. The waterlogged images were captured by simulating blockage events where water pooled around the drain guard. The non-waterlogged images were recorded during dry conditions and normal rainfall flow. All captured images were utilized for the classification task. The dataset was partitioned in a 70%-15%-15% ratio for training, testing, and validation purposes, respectively, to ensure robust model evaluation.

3.1 Data Preprocessing

To ensure compatibility with the InceptionV3 architecture, all input images were initially resized to 224 x 224 dimensions. The pixel intensity values were normalized by dividing them by 255, scaling the range between 0 and 1 to facilitate faster convergence during training. Unlike standard central cropping, which risks removing peripheral drainage details, we resized the full region of interest to preserve context. We also shuffled the images within their respective classes to prevent the model from learning patterns based on the sequence of data capture.

For the training phase, the data was divided into batches, with a batch size of 32. To combat overfitting caused by the limited size of our custom dataset, extensive data augmentation techniques were applied. Images were rotated randomly by up to 20 degrees and shifted along their width and height to simulate camera vibrations or installation variances. Additionally, random zooming and shearing transformations were introduced, along with horizontal flipping, to ensure the model remains robust against varying environmental conditions.

3.2 InceptionV3

InceptionV3 is a Convolutional Neural Network (CNN) based architecture known for its efficiency and firm performance in image classification tasks. Its wide application range and robust feature extraction capabilities make it highly suitable for waterlogging detection tasks. Convolutional kernels of various sizes are utilized by InceptionV3, which permits its receptive fields to feature different scopes. It uses a modular approach and final joining to reduce the network's design area while learning the fusion of features with different scales. Table 2 presents the Network Structure of the InceptionV3 model.

A Batch Normalization (BN) layer has been integrated into the InceptionV3 architecture as a regularizer between the fully connected (FC) layer and the auxiliary classifier. The batch gradient descent approach was implemented into the BN model to accelerate the deep neural network's training speed and convergence.

$$B = \{x_{1...m}\}, \gamma$$

$$\{y_i = BN_{\gamma, \beta}(x_i)\}$$

$$\mu_B \leftarrow \frac{1}{m} \sum_{i=1}^m X_i$$

$$\sigma_B^2 \leftarrow \frac{1}{m} \sum_{i=1}^m (X_i - \mu_B)^2$$

$$\hat{x}_i \leftarrow \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$$

$$y_i \leftarrow \gamma \hat{x}_i + \beta \equiv BN_{\gamma, \beta}(X_i)$$

Here, γ and β are learnable parameters, x represents the activation value of batch B , and m is the number of activation values. The variation in the value distribution is adjusted by γ , and the average value's position is adjusted by β . The average value in one dimension is represented by μ_B , the variance in each feature map dimension is denoted by σ_B , and ϵ is a constant.

3.3 Our Transfer Learning-based Model

Research indicates that the InceptionV3 model effectively performs image recognition and classification tasks by modifying the design of the fully connected layers while reserving the parameters of the convolutional layers.

Given that our custom dataset is limited—containing approximately 450 images—training a deep learning model from scratch is computationally inefficient and prone to overfitting. To address this challenge, we utilized a pre-trained model trained on the vast ImageNet dataset. ImageNet contains approximately 14 million images, providing a robust baseline for visual feature recognition⁴. Consequently, we employed the transfer learning technique, leveraging the ImageNet weights learned by the pre-trained model.

Regarding fine-tuning, we extracted features using the convolutional layers of the pre-trained InceptionV3 model as the foundation. This approach allows our system to inherit powerful feature extraction capabilities. Our proposed layer architecture is illustrated in Figure 1.

As we adjusted the input parameters and size of the feature maps to 224x224x3, corresponding modifications were made to the network architecture structure. The initial modified parameters are presented in Table 3. Following all structural adjustments, the final model parameters are detailed in Table 4.

3.4 Hyper-parameter Settings

For our rooftop waterlogging detection model, we utilize specific hyper-parameter settings to optimize performance. The activation function used is Sigmoid, which is effective for binary classification tasks (differentiating between blocked and clear drains). We employ the Adam optimizer with a learning rate of 0.001, balancing convergence speed and accuracy.

The loss function selected is Binary Cross entropy, which is suitable for our binary classification problem. We set the batch size to 32, allowing for efficient training and stable gradient

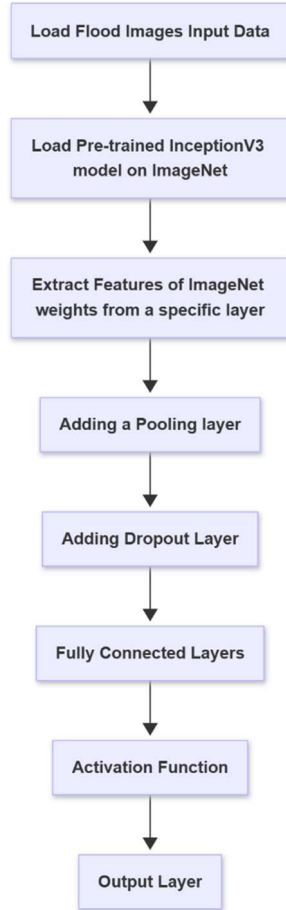


Fig. 1: Our proposed layer architecture

Table 2: Network Structure of our Modified Inception V3 Model

Type	Patch Size/Stride	Input Size
Conv	3x3 / 2	224x224x3
Conv	3x3 / 1	111x111x32
Conv padded	3x3 / 1	109x109x32
Pool	3x3 / 2	109x109x64
Conv	3x3 / 1	224x224x80
Conv	3x3 / 2	224x224x192
Conv	3x3 / 1	25x25x288
InceptionA x 3		32x25x288
InceptionB x 5		12x12x768
InceptionC x 2		5x5x1280
Fc	51200x1024	5x5x2048

updates. The model was trained over 20 epochs to ensure sufficient learning of the drainage features while avoiding overfitting. These hyper-parameters collectively enhance the model's ability to detect drain blockages accurately.

3.5 VGG-16

Characterized by its simplicity and consistent design, VGG-16 uses stacked convolutional layers with modest receptive fields. The model's clear design enables analyzing its performance and contributions to the goal. In the field of deep learning architectures, the VGG16 model serves as an iconic depiction of simplicity and efficacy. Specifically developed for image classification tasks, its unique structure incorporates stacks of convolutional layers, providing the path for thorough feature extraction. This investigation aims to untangle the deep aspects of the VGG16 model adapted for the tough job of rooftop waterlogging detection. Each of the five convolutional blocks that make up VGG16 contains multiple convolutional layers, which are subsequently followed by a max-pooling layer.

3.6 ResNet-50

A version of the ResNet design, ResNet-50 incorporates skip links to avoid the vanishing gradient issue. These skip connections improve the training of deep networks, making ResNet-50 a popular option for image categorization. The ResNet50 model stands as a peak in the area of convolutional neural networks (CNNs), especially famous for its ability to tackle the problems of training extremely deep networks.

Developed by Microsoft Research, it introduced the notion of residual learning, allowing for the building of deeper networks without the problem of vanishing gradients. In this research, we deconstruct the structure of the ResNet50 model suited for the detection of rooftop waterlogging using surveillance data. The ResNet50 design contains 50 layers, emphasizing a modular and hierarchical structure. The model is structured into phases, each having convolutional blocks. These blocks are further divided into identity blocks and convolutional blocks, creating the backbone of the network.

3.7 MobileNet

MobileNet is a Convolutional Neural Network (CNN) design tailored specifically for efficient image classification on mobile and embedded devices with constrained computational resources. Created by Google, MobileNet achieves high classification accuracy while maintaining a compact model size and low computational complexity. At the core of MobileNet lies the depthwise separable convolution operation, which consists of two distinct convolutional steps: depthwise convolution and pointwise convolution³. While the pointwise convolution (often referred to as point-by-point) combines the results with a 1 x 1 convolution to form the final feature map, the depthwise convolution applies a single filter to each input channel independently. This separation significantly reduces the number of parameters and computations compared to traditional convolutional layers, making MobileNet lightweight and efficient.

4. Result and Discussion

A transfer learning-based classification model was developed in the study to detect waterlogging. The model utilized various base models such as VGG16, ResNet50, MobileNet,

and InceptionV3. Among these models, InceptionV3 and MobileNet demonstrated superior performance, which led to further fine-tuning.

4.1 Model Training Performance

We are working on four models: InceptionV3, MobileNet, VGG16, and ResNet-50. In the initial training phase, both VGG16 and ResNet-50 demonstrated moderate performance, each achieving an accuracy of 87.5%. In contrast, the MobileNet and InceptionV3 base models performed significantly better, both reaching a superior accuracy of 93.8%. These two models showed the highest effectiveness among all tested architectures during learning from the training dataset. The visual representation of these base model accuracies is shown in Figure 2: Accuracy of Base Models. Based on these initial evaluations, both MobileNet and InceptionV3 have been identified as the most promising candidates for further development and fine-tuning.

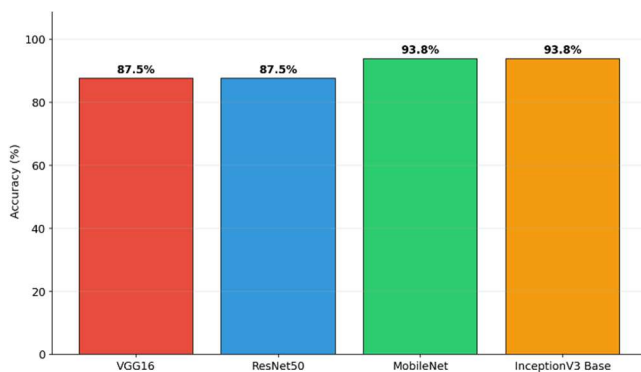


Fig. 2: Accuracy of Base Models

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